

Program: Diploma in Electrical Engineering & Electric Vehicle Technology	
Course Code: 5418	Course Title: Electric Vehicle Testing and Circuits Lab
Semester: 5	Credits: 1.5
Course Category: Program Elective Course	
Periods per week: 3(L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To compare different EV configurations
- To familiarize wiring circuits of EV
- To apply wheel balancing and wheel alignment of EV
- To apply the procedures of hydraulic brake bleeding and disc brake overhauling, Rectifier, inverters and testing of EV batteries

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge about the elements of Electrical Engineering, EV technologies		Fundamentals of Electrical and Electronics	2

Course Outcomes:

On completion of the course ,the students will be able to:-

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Identify the comparisons between electric and hybrid vehicles	7	Applying
CO2	Select various wiring circuits of Electric Vehicle	8	Applying
CO3	Make use of Wheel balancing machine, Computerized wheel alignment	9	Applying

CO4	Apply procedures for hydraulic brake bleeding and disc brake overhauling, Rectifier ,inverters and testing of EV batteries	15	Applying
	Lab Exam	6	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3			3			
CO3	3			3			
CO4	3			3			

3-Stronglymapped, 2-Moderatelymapped,1-Weaklymapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Identify the comparisons between electric and hybrid vehicles		
M1.01	Demonstrate the power train layout of electric and hybrid vehicles	2	Understanding
M1.02	Develop a comparison chart of petrol, diesel, electric and hybrid vehicles	2	Applying
M1.03	Demonstrate the operation of regenerative braking	3	Applying
CO2	Select various wiring circuits of Electric Vehicle		
M2.01	Identify wiring circuits in various electric vehicles including wiring harness, color codes and cable specifications	2	Understanding
M2.02	Construct wiring of headlight circuit	2	Applying

M2.03	Build the wiring of horn circuit and side indicator	2	Applying
M2.04	Build the wiring of EV batteries, BMS wirings.	2	Applying
CO3	Make use of Wheel balancing machine, Computerized wheel alignment machine, Automatic Tyre changer		
M3.01	Make use of wheel balancing procedure.	4	Applying
M3.02	Make use of Tyre changing procedure on pneumatic tyre changer.	3	Applying
M3.03	Experiment with computerized wheel alignment.	2	Applying
CO4	Apply procedures for hydraulic brake bleeding and disc brake overhauling, Rectifier ,inverters and testing of EV batteries		
M4.01	Apply procedures for hydraulic brake bleeding and disc brake overhauling	5	Applying
M4.02	Demonstrate the operation of Rectifiers, Inverters, DC/DC converters and Power Split devices for Hybrid Vehicles	5	Applying
M4.03	Apply the procedures of testing on batteries	5	Applying

Student Activity

Suggested Open-ended Experiments:

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open ended experiments as a group of 2-3. There is no duplication in experiments between groups.

1. Convert a petrol engine two wheeler into an electric two wheeler using a hub motor and control units
2. Construct a comparative performance chart of different batteries used in electric vehicles today.

Text/Reference:

T/R	Book Title/Author
T1	Laboratory Manuel on Electrical Measurements & Electrical Machines, P.O. Kuttappan, Spades publishers
R1	Electrical Measurements & measuring instruments by A.K. Sawhney Dhanpat Rai Publications
R2	Electrical Measurements & measuring instruments by R.K. .Rajput, S.CHAND Publications
R3	Electronic Measurements & instrumentation by R.K.Rajput, S.CHAND Publications
R4	A.K. Babu, Electric & Hybrid Vehicles, Khanna Publishing House, New Delhi Ed.
R5	T. Denton, “Electric and Hybrid Vehicles”, Routledge
R4	Electrical Measurements & measuring instruments By G.K. Banerjee PHI Publications

Online Resources:

Sl.No	Website Link
1	Virtual Lab: http://vlabs.iitkgp.ernet.in/asnm/#
2	https://nptel.ac.in/course.html
3	https://www.electrical4u.com/electrical-engineering-articles/measurement/
4	www.youtube.com
5	https://www.allaboutcircuits.com/

Student Activity**Suggested Open-ended Experiments:**

Students can do open ended experiments as a group of 3-5. There is no duplication in experiments between groups. This is mainly for the purpose of continuous internal evaluation and a score of 15 marks. Students should prepare a separate report on the open ended experiment of their choice.

Example:1. Calibrate the instruments in the laboratory and identify the error chart.